

Planning Context and Proposed Ohio River South Shore Trail (ORSST) Alignment

Moon Township has approximately 2.8 miles of Ohio River frontage, all within Section 500.

History and Cultural Resources

Moon Township has evolved significantly since its beginnings as a farm-based community. As the oldest township in Allegheny County, founded in 1788, Moon Township had a total area of 143 square miles. Moon Township is named from the crescent-like bend of the Ohio River on its northern boundary. Some reports indicate that it would take one man on horseback two days to travel from one end of the community to the other. This geographically large township eventually spawned into 55 smaller municipalities, including the current neighboring townships of Fayette, Findlay, Crescent and the Borough of Coraopolis.

In its early days, settlers in Moon Township depended heavily on the hunting and farming economy for survival. The excess of farm production at the end of the 18th century brought about a need for industries such as gristmills and sawmills. The Township continued to experience significant economic growth into the 20th century when roadways and railroads opened up the gates to Moon, making it an attractive place for people to settle and raise their families. Both the Sewickley Bridge, which was originally constructed in 1911, and the Pennsylvania & Lake Erie (P&LE) Railroad contributed significantly to Moon Township’s tremendous population growth.

During World War II, the industrial plants located within Moon Township and the surrounding communities became major suppliers of armor plates and munitions. This boom in production created a great housing need for workers who were stationed at factories in Neville Island and along the Ohio River.

Moon Township’s largest percentage of growth came in 1952, when the Greater Pittsburgh Airport was completed, dedicated and opened for business in Moon Township. One year later, construction on the Penn Lincoln Parkway was complete, making the commute to downtown Pittsburgh from Moon Township possible in about 20 minutes. These two large development projects not only contributed to a population increase of 24 percent between 1950 and 1957, but also led to a housing boom that created more than 1,250 homes in a decade-long time span.

The construction of the Greater Pittsburgh Airport, which was later named Pittsburgh International Airport, is perhaps the most significant contributing factor to the economic growth Moon Township experienced. The growth was challenged in the early 1990s, however, when the airport relocated to Findlay Township. At that time, the Moon Township Board of Supervisors developed a forward-thinking plan that would help sustain growth in the Township. Although Moon lost a great source for business development, the Township has continued to experience growth and today is home to a number of high-profile national corporations, such as FedEx Ground and GlaxoSmithKline. In addition, Moon Township also serves as the home of Robert Morris University, which has a very strong business curriculum and educates nearly 5,100 students annually.

Within the portions of Moon Township located in the ORSST study area there are no sites on the National Register of Historic Places or locally designated sites.

Land Use, Parks and Recreation Resources and River Access

The land use within the Township’s portion of the ORSST study area is mostly industrial or major roadway infrastructure. The stretch of riverfront in this area is predominately oil tank farms used for the storage and distribution of various commercial and residential petroleum products.

The Township has more than a dozen dedicated parks and recreation sites and facilities ranging in size from tot lots to the 236 acre Moon Park. None of the Township’s park and recreation facilities are located within the ORSST study area. The Township has, however, acquired the former RB&W Bolt Plant; a brownfields site located along the Ohio Riverfront. The Township is intending on creating a new waterfront park on this and adjacent parcels. The Township is in the process of remediating the environmental issues on the site in compliance with the Pennsylvania Department of Environmental Protection’s Voluntary Cleanup Program. The Township has applied for grant funding from DCNR to support a master planning effort for this park. This proposed park and recreation facility along with the proposed Ohio River Trail, will create a new 2.5 mile public stretch of Ohio River frontage, where none exists today.

Moon Township is fortunate to contain a major portion of the Montour Trail. This 47 mile long trail connects 14 municipalities through southwestern Pennsylvania. Future trail plans anticipate that the Montour Trail will connect directly to downtown Pittsburgh and to Washington, D.C. via the Great Allegheny Passage.

The Township adopted a Sidewalks and Trails Master Plan in 2006. This plan, along with the Township’s 2000 Comprehensive Plan and the 2005 Comprehensive Parks, Recreation and Open Space Plan, provides a comprehensive analysis of the existing sidewalk and trails network. It also identifies areas where existing infrastructure needs to be upgraded and where new infrastructure is needed. The plan makes the following specific recommendations:

- *Complete missing gaps in the existing sidewalk network to greatly enhance interconnectivity;*
- *Connect new trails and sidewalks to the Montour Trail;*
- *Integrate the riverfront area into the community-wide trail system and the Montour Trail; and*
- *Develop new transportation, urban design and infrastructure strategies into improvement plans for the University Boulevard Corridor.*

Section 500 - Moon Township

Community Character and Demographic Overview

Moon Township is a large suburban township located 20 Miles west of downtown Pittsburgh and is bordered by the Ohio River, Coraopolis Borough and Crescent Township on the North and Robinson Township on the east. Today, Moon Township is approximately 23 square miles following the creation of seventeen boroughs and townships from the original 143 square miles size of the original Moon Township.

Moon Township’s population has been steadily growing since 1960. This growth can be attributed in large part to the construction of the Pittsburgh International Airport in 1952. The last four decades have brought a lot of change to Moon Township. It was once a rural farming community and it is now a thriving suburban township with a large population. Moon Township has developed into a diverse community that offers many different types of housing and the Township has extensive commercial areas along with green spaces and is in close proximity to major regional transportation infrastructure.

Moon Township consists of approximately 22,290 people and 8,445 households. The racial makeup of the area is 93.2 percent White, 3.6 percent African American, 0.99 percent Hispanic, 0.06 percent Native American, and 1.9 percent Asian. The per capital income of the town is \$26,457 and 2.2 percent of the population live below the poverty line, significantly lower than Pennsylvania’s rate of 12.1 percent.

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The community survey results taken during the preparation of the 2005 Comprehensive Recreation, Parks and Open Space Plan indicated that 66.7 percent of the respondents felt that developed trail are very important or important. A follow up survey performed for the preparation of the 2006 Sidewalks and Trails Master Plan supported this previous result with 79 percent of the respondents reporting that a connected system of sidewalks and trails is either very important or important.

Environmental Context and Infrastructure

River's Edge and Habitat: The Ohio River's edge in Moon Township has some unique qualities when compared to other stretches within the ORSST study area. The area of the proposed waterfront park and the Moon Township Authority's supply wells consists of a relatively gradual slope from the Ohio River's mean water elevation to the CSX Railroad right-of-way. Some of these areas exhibit characteristics of wooded wetlands, with larger canopy trees and signs of periodic inundation. This is a valuable habitat that could be enhanced as part of a new park landscape.

The Ohio River's edge between the Sewickley Bridge and the U.S. Army Corps of Engineers Dashields Lock Complex exhibits steep topography and the area of the former P&LE Stoops Ferry train station exhibit very steep slopes. There is rich and dense vegetative cover in this area, most likely since it has experienced minimal disturbance beyond the impacts of major flood events, since the construction of the railroad in the 1870s.

Streams and Stormwater:

Narrows Run is a major day-lighted stream which travels down a steep valley, parallel to University Boulevard. The stream passes underneath University Boulevard/PA Route 51 and the CSX Railroad right-of-way via a long culvert. This culvert outfall is located near the former Stoops Ferry train station area of the Ohio River further complicating the ability to locate a trail in this area.

Thorn Run is a stream course that travels down a valley located along the perimeter of Coraopolis Borough. It travels underneath PA Route 51 near its intersection with Thorn Run Road. The stream passes through several box culverts to get to the north side of PA Route 51 near American Bridge Way. At this point the stream appears to pass underneath an old warehouse building and a parking lot. It then travels through a newer box culvert underneath the CSX Railroad right-of-way. Next it travels through the Petroleum Products tank farm site in a culvert or rip-rap channel until it discharges into the Ohio River.

Transportation Infrastructure

PA Route 51 in Section 500 is a very wide roadway with a median. It is a higher speed roadway with a limit-access design condition between the intersection of University Boulevard/Stoops Ferry Road and Thorn Run Road. PA Route 51 in Section 500 from Stoops Ferry Road to Thorn Run

has an average daily traffic volume of approximately 22,000. This high volume is partly a function of the convergence of several major arterial roadways, as well as traffic generated by the access to the Sewickley Bridge. The roadway has shoulders along most of this segment.



High Speed Environment along PA Route 51 in Moon Township

Economic Development

Since its inception in 1954, the Pittsburgh International Airport has played an important role in the economic growth of Moon Township. One area of Moon Township that is seeing a great deal of renewed commercial development is the University Boulevard Corridor, which was previously known as Beers School Road and Narrows Run Road. With the opening of numerous restaurants, hotels and other airport-related businesses, University Boulevard began to see a significant increase in traffic. This growth continued until 1992 when the Pittsburgh International Airport terminal relocated to the adjacent township of Findlay, taking with it a great deal of airport-related traffic and thus altering the boulevard's identity as an airport service corridor. Today, those businesses have developed to reflect the needs of Moon Township's corporate businesses, office parks and Robert Morris University's student body.

In order to be proactive and abate the consequences of Pittsburgh International Airport's relocation, township officials developed a forward-thinking plan that would maintain the corridor's commercial success. The first step in ensuring stability was to develop a strategic plan that would guide the future growth and sustainability of development in this thoroughfare. The key action identifying this strategic plan was the renaming of Beers School Road/Narrows Run Road to University Boulevard, a name that reflect the road's main anchor, Robert Morris University. This plan also included a conceptual design for improvements to University Boulevard to improve traffic flow, enhance pedestrian access and mobility and develop streetscape improvements. In addition, the plan recommended how to implement urban design changes to the corridor such as landscaping, sidewalks, building facades, public amenities and a gateway. The plan was presented and approved by the Moon Township Board of Supervisors in 2003.

Since that time, the plan has been further enhanced with the approval of an overlay district, designed as a tool to implement the recommendations included in the strategic plan. Over the past five years, vast improvements have been made to University Boulevard. Nearly \$25 million in developments have been added, including a Double Tree Hotel, Sheetz convenience store and a Primanti Brothers Restaurant, one of Pittsburgh's cultural icons. Robert Morris University has also enhanced the beauty and appearance of their campus with the addition of an iron archway entrance and a multi-million dollar football stadium, which sits above University Boulevard. Moon Township is now also home to a 14,000 square-foot Walgreens pharmacy/drug store. Additionally, Wal-Mart Corporation has also submitted plans to build a 150,000 square-foot supercenter where the West Hills Shopping Plaza currently exists. Those plans are currently being reviewed by the Moon Township Board of Supervisors.

The redevelopment along University Boulevard and the continual addition of new businesses allows Moon Township officials to continue to pursue funding for the improvements that are outlined in the plan.

Proposed ORSST Route and Alternatives

Proposed ORSST Route Description through Section 500

The proposed route through Section 500 would travel from Shouse Park in Crescent Township to Thorn Run Road in Moon Township via a 10' wide path through the U.S. Army Corps of Engineers (ACOE) Dashields Lock Complex, CSX and Moon Township properties and then via a Duquesne Light utility corridor to Moon Township/Coraopolis Borough Municipal Line near Thorn Run.

Route Characteristics and Issues

U.S. Army Corps of Engineers Dashields Lock Area: The ORSST route through Section 500 would consist of a continuous 10' wide side path along the CSX Railroad side of the ACOE Dashields Lock access roadway and a 10' wide multi-use trail (including a boardwalk portion) along the river side of ACOE and CSX Railroad right-of-way to the Moon Township Authority's riverfront parcels.

Several conversations were held with representatives of the ACOE as part of the ORSST study effort, including with the Project Manager of the Upper Ohio River Navigation Study. The ACOE is currently studying improvement alternatives to the Dashields Lock and environs as part of the larger Navigation Study. Discussions were also held with the ACOE Reality Specialist involved with determining ACOE property ownership and related land matters. Based on feedback provided by the ACOE, the possibility of extending the trail along the Dashields Lock property exists, but will require a formal request to the ACOE along with ongoing coordination.

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Army Corps of Engineers Dashields Lock Area adjacent to the CSX Railroad

Stoops Ferry Area: The existing topography along the river in the Stoops Ferry area consists of a narrow shelf wide enough to accommodate the railroad right-of-way; there is limited space for additional facilities. A boardwalk will be required in the area of Stoops Ferry to provide for the width of the trail and to achieve the required setback from the railroad. It is anticipated that the boardwalk will need to be approximately 2,270 feet (4 of mile) in length in this area. CSX Railroad right-of-way will need to be acquired in the area of Stoops Ferry where CSX's right-of-way extends beyond its typical width to the Ohio River's edge. The construction of the boardwalk will likely be handled via barge from the river in order to eliminate or minimize any operational impact to railroad operations.



CSX Railroad with Former Stoops Ferry Station Platforms Visible

Moon Township Riverfront Area to Thorn Run: The Moon Township Authority's property includes land utilized by the Authority for water supply wells and a former brownfields parcel targeted for a new waterfront park by the Moon Township Department of Parks and Recreation. The Authority also owns, or is in the process of acquiring, several other parcels to the west of the proposed park parcel. From this area the proposed trail would extend from the western parcel boundary, parallel to the river and the CSX Railroad right-of-way to the proposed park parcel. The trail is proposed to be a 10' wide gravel path in this segment. A portion of the trail through the proposed park property would be located on a concrete building slab from the former industrial building where a railroad siding once entered the industrial facility. The trail is proposed to be asphalt in the location of the concrete slab to the at-grade railroad crossing at Valley Ambulance Drive.

The trail will cross the existing at-grade railroad crossing and then travel parallel to the CSX Railroad on a 30' wide utility corridor right-of-way from Valley Ambulance Drive to 4th Avenue, just west of Thorn Run. A bridge span or culvert will be required to cross Thorn Run, behind the PennDOT guidrail, in this location.

In the area between Valley Ambulance Drive and Thorn Run, the ability to acquire an easement on the 30' wide Duquesne Light utility corridor will need to be determined including addressing existing landscape/nursery operation located across the utility right-of-way in the area of Thorn Run Road. It is not know whether this operation has an agreement with Duquesne Light or if it is an encroachment.

There will need to be utility relocation coordination near the intersection of 4th Avenue and American Bridge Way/Tri State Hose & Supply as well as the creation of a structure to cross Thorn Run.

All of the trail segments in Section 500 that are located immediately adjacent to the CSX Railroad right-of-way are proposed to be fenced.



Sewickley Bridge with Future Moon Township Parkland Parcel in Foreground

Proposed Route Lengths

Total Length of On-Road Route through Crescent Township (Section 500) = .2 miles

Moon Township 10' Wide Multi Use Path = 2.1 miles

Moon Township 10' Wide Multi Use Path on Boardwalk = .4 miles

Total Length of Route through Moon Township = 2.5 miles

Identified ORSST Routing Alternatives in Section 500

Alternative 500A

Description: This alternative is linked to Alternative 400E and consists of the current Pennsylvania Bicycle Route A, which travels on PA Route 51/ Stoops Ferry Road/University Boulevard from the Section 400 section line to Valley Ambulance Drive.

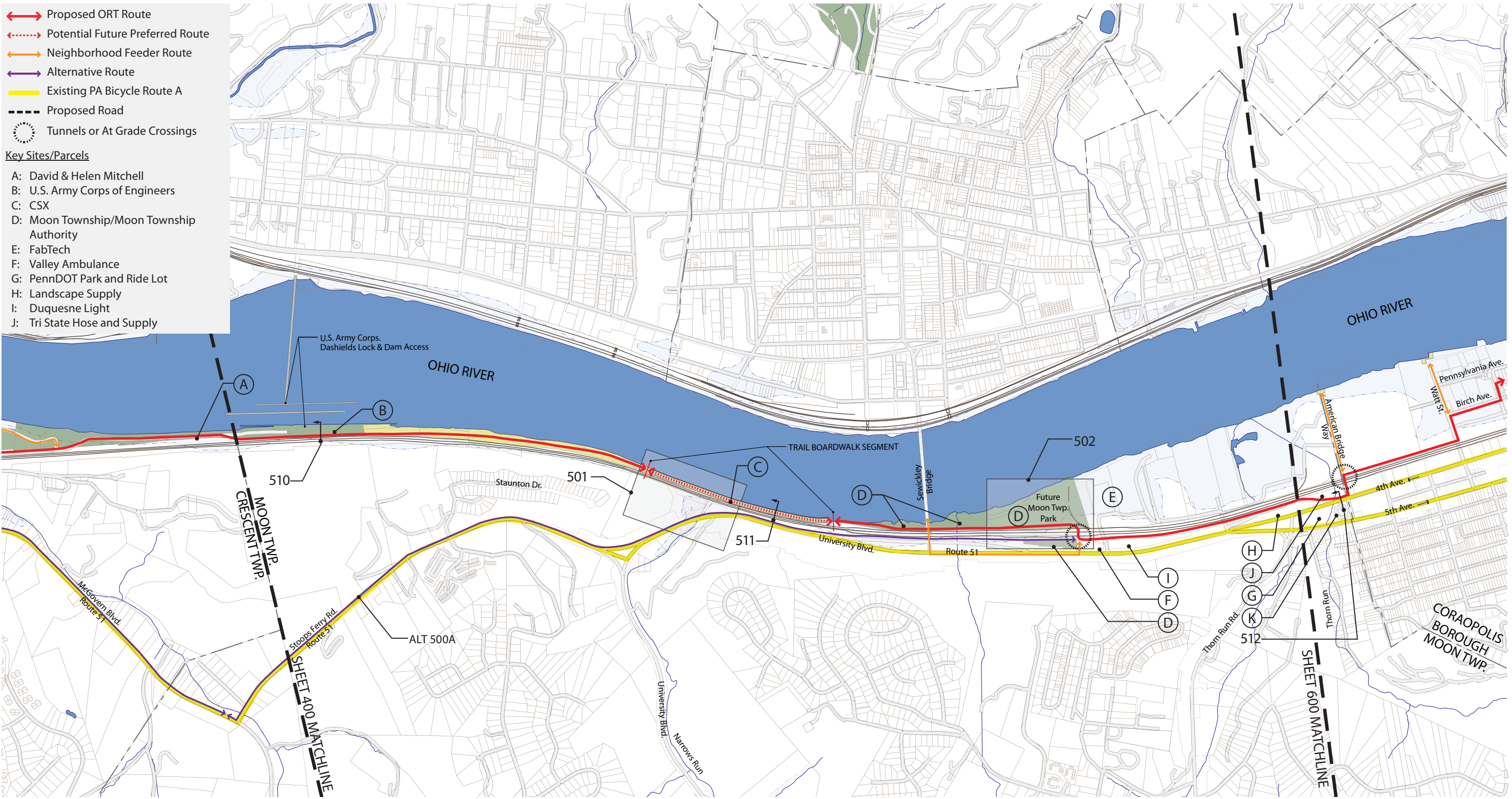
Issues and Opportunities: This segment of PA Route 51 has the highest traffic volumes within ORSST study area. Stoops Ferry Road has an average daily traffic volume of approximately 14,000 and University Boulevard has an average daily traffic volume of approximately 22,000. Stoops Ferry Road also has a significant topographic change that would create a major barrier for recreational bicyclists to climb. Stoops Ferry Road does have 10' wide (average) shoulders on each side. University Boulevard is a very wide, four-lane highway designed with a limited access character and is not a pedestrian or bicycle friendly setting.

LEGEND

- Proposed ORT Route
- Potential Future Preferred Route
- Neighborhood Feeder Route
- Alternative Route
- Existing PA Bicycle Route A
- Proposed Road
- Tunnels or At Grade Crossings

Key Sites/Parcels

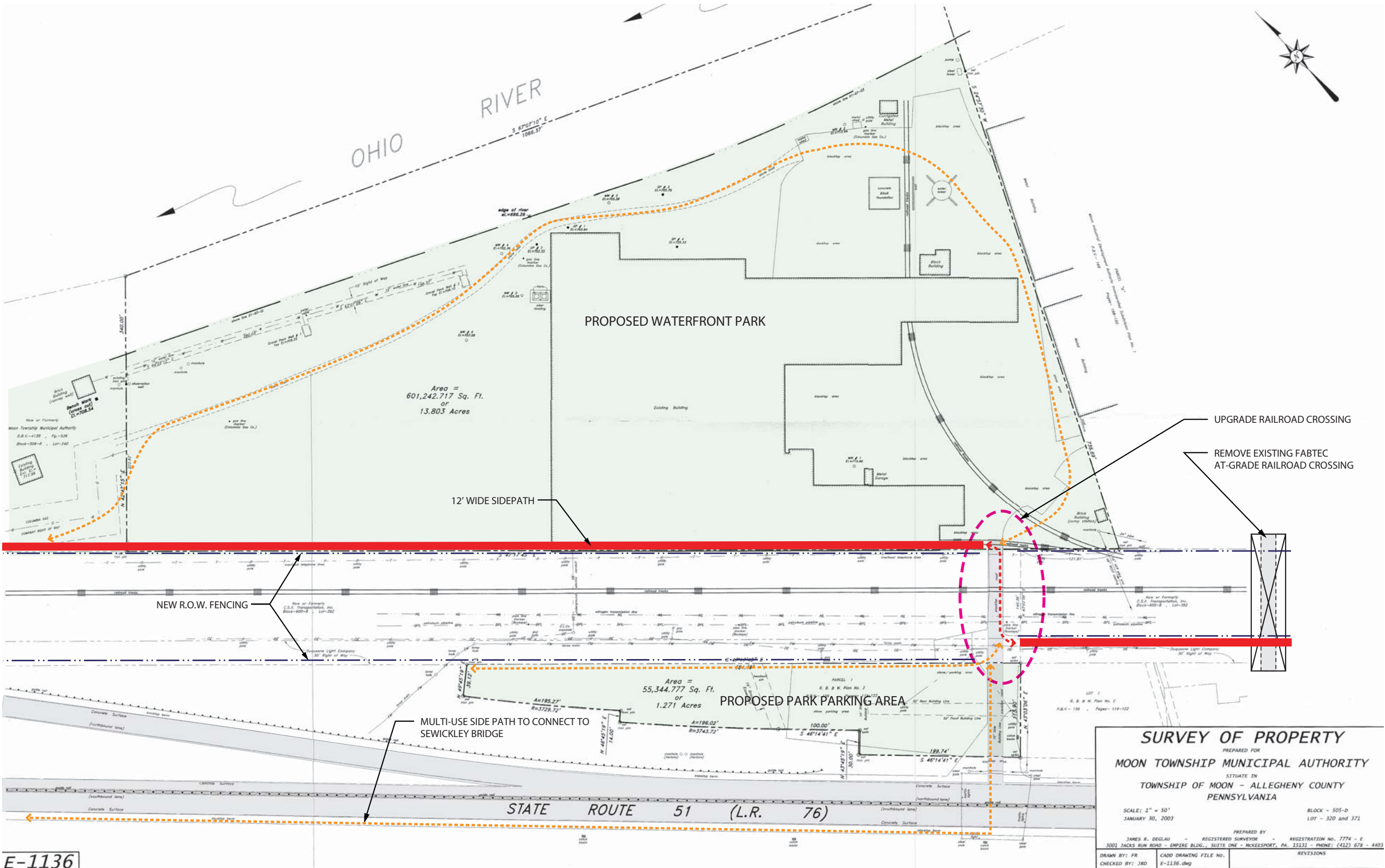
- A: David & Helen Mitchell
- B: U.S. Army Corps of Engineers
- C: CSX
- D: Moon Township/Moon Township Authority
- E: FabTech
- F: Valley Ambulance
- G: PennDOT Park and Ride Lot
- H: Landscape Supply
- I: Duquesne Light
- J: Tri State Hose and Supply



LEGEND

- Signed On-Road Route
- Neighborhood Feeder Route
- Off-Road Route
- Trail Blazer Sign

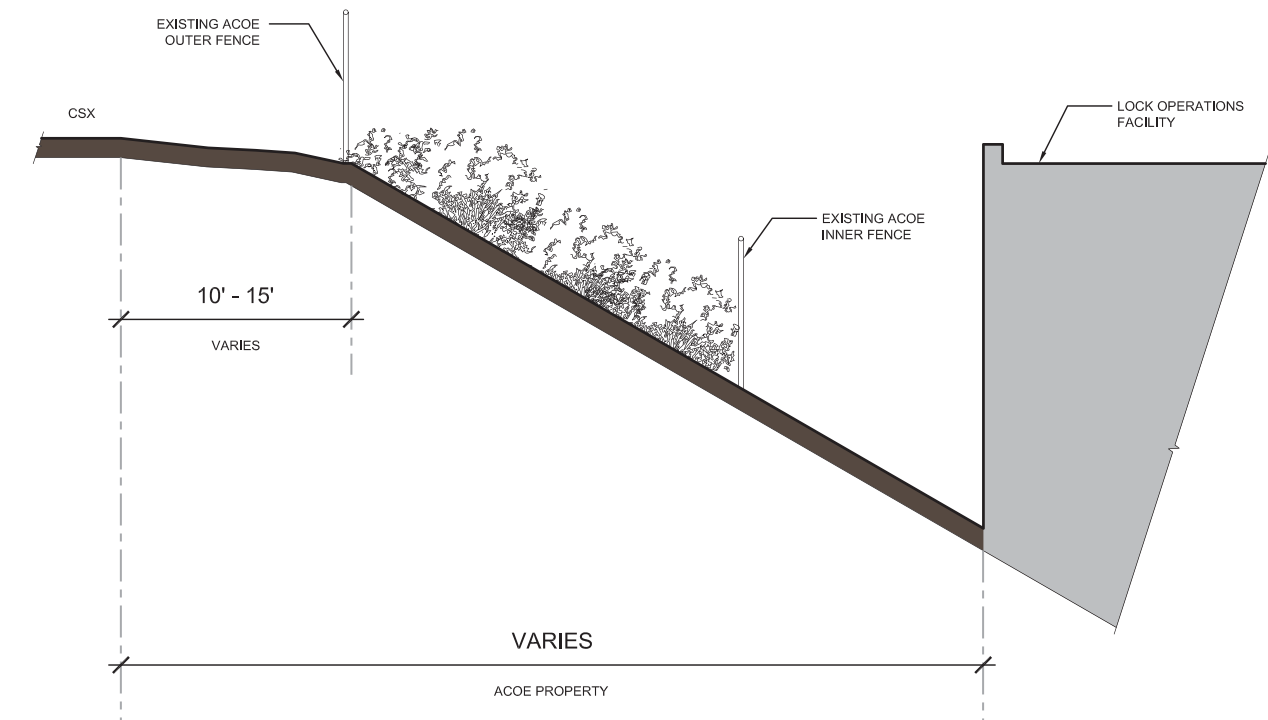
Moon Township Municipal Authority Property - 502



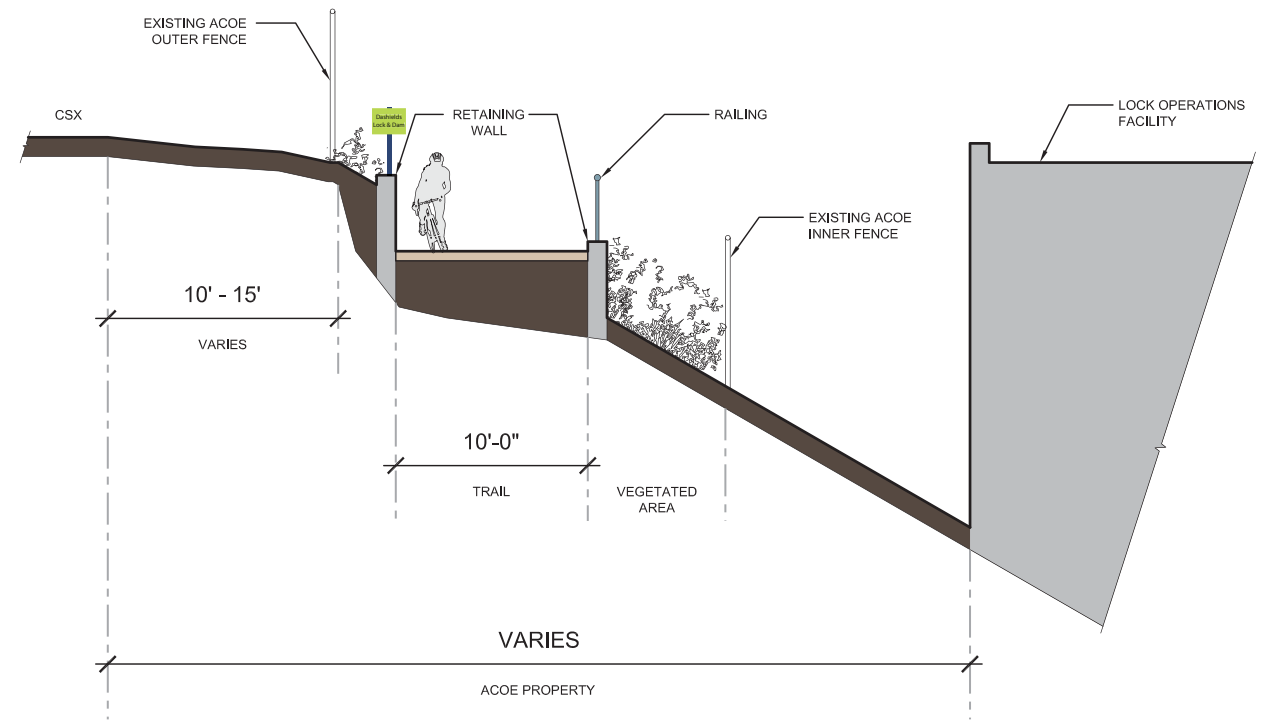
E-1136

502

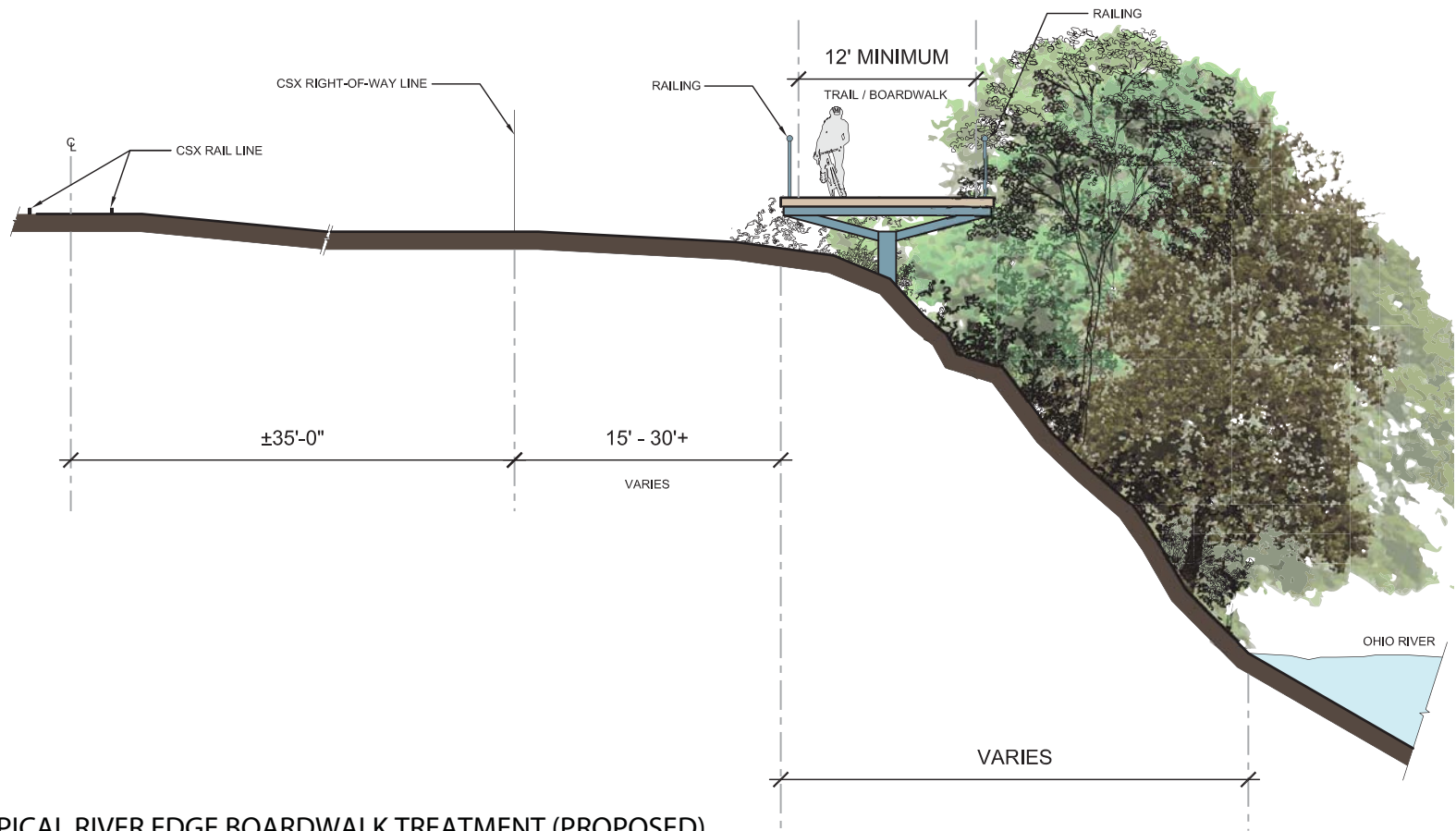
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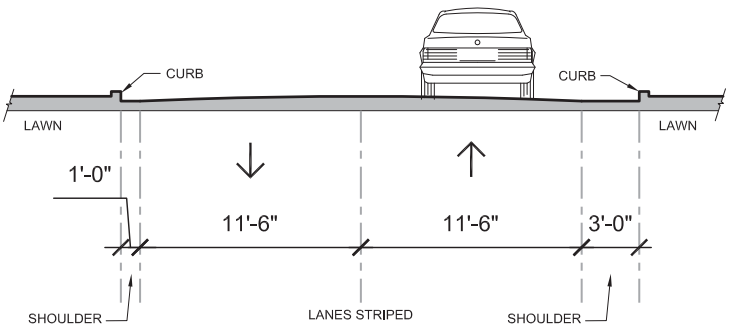
510 ACOE DASHIELDS LOCK AREA (EXISTING)
SCALE: 1"=10'



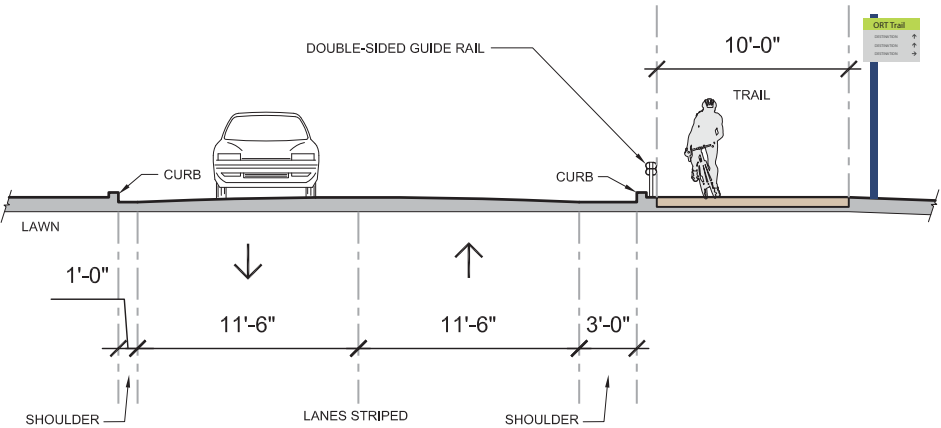
510 ACOE DASHIELDS LOCK AREA (PROPOSED)
SCALE: 1"=10'



511 TYPICAL RIVER EDGE BOARDWALK TREATMENT (PROPOSED)
SCALE: 1"=10'



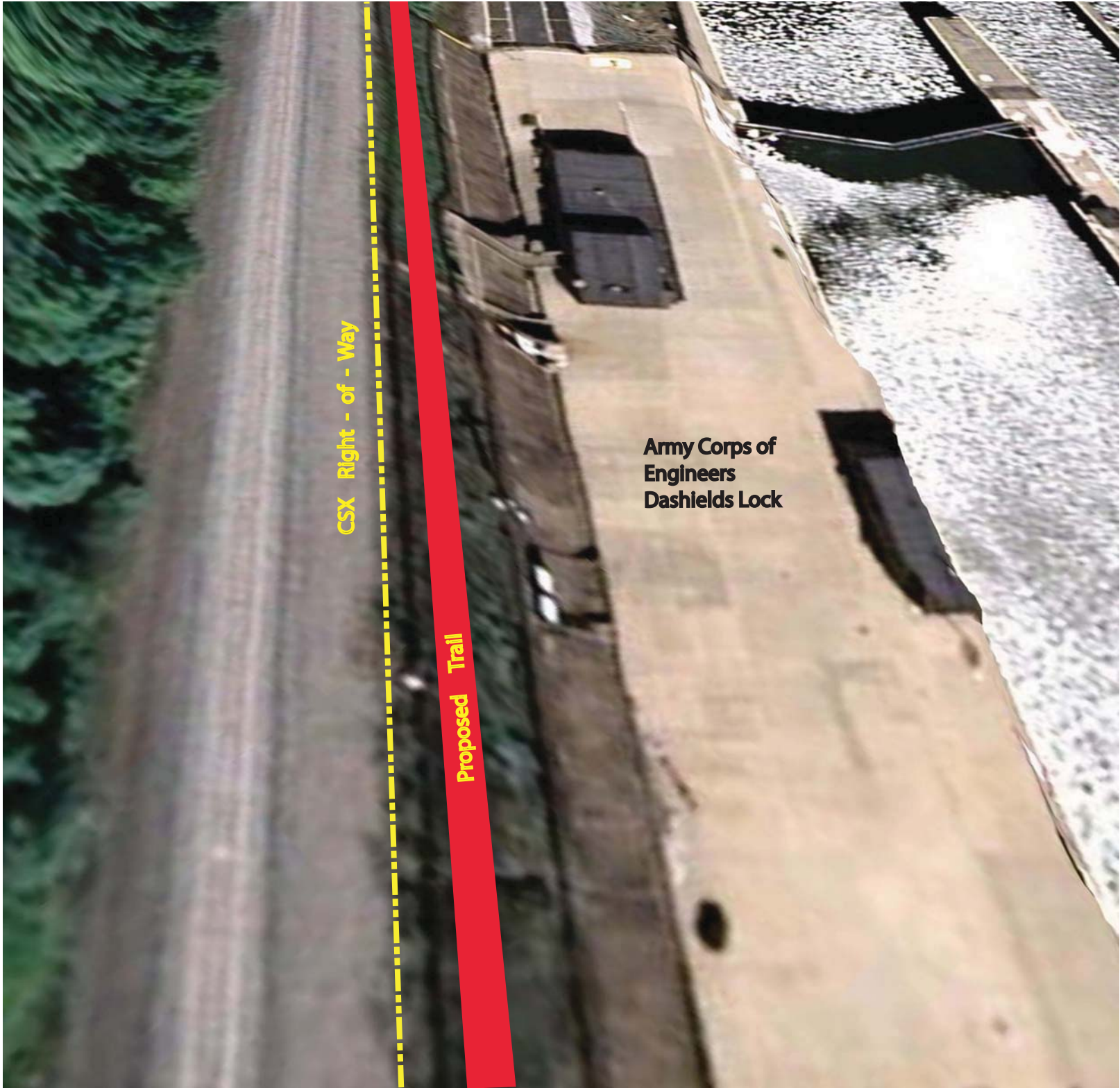
512 MOON TWP. 4TH AVENUE IN THE VICINITY OF THORN RUN LOOKING WEST (EXISTING)
SCALE: 1"=10'



512 MOON TWP. 4TH AVENUE IN THE VICINITY OF THORN RUN LOOKING WEST (PROPOSED)
SCALE: 1"=10'

LEGEND

Proposed 10' Wide Multi-Use Path Trail Route



CSX Right-of-Way in the Vicinity of Dashields Lock



Site Photo Location

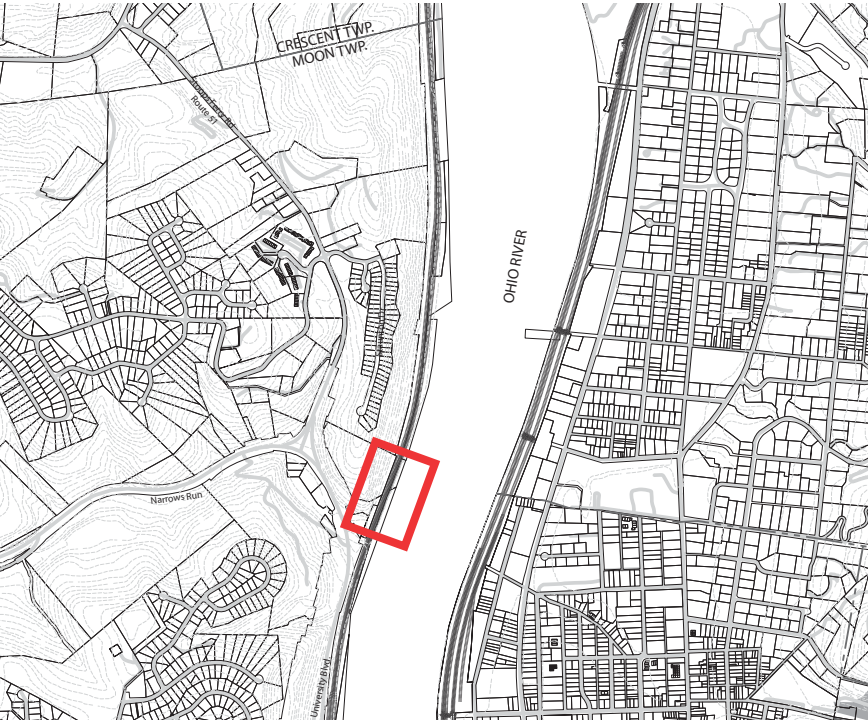
520 Proposed Trail Location
SCALE: Not to Scale

LEGEND

Proposed 10' Wide Multi-Use Path Trail Route



CSX Right-of-Way in the Vicinity of Former P&LE Stoops Ferry Station

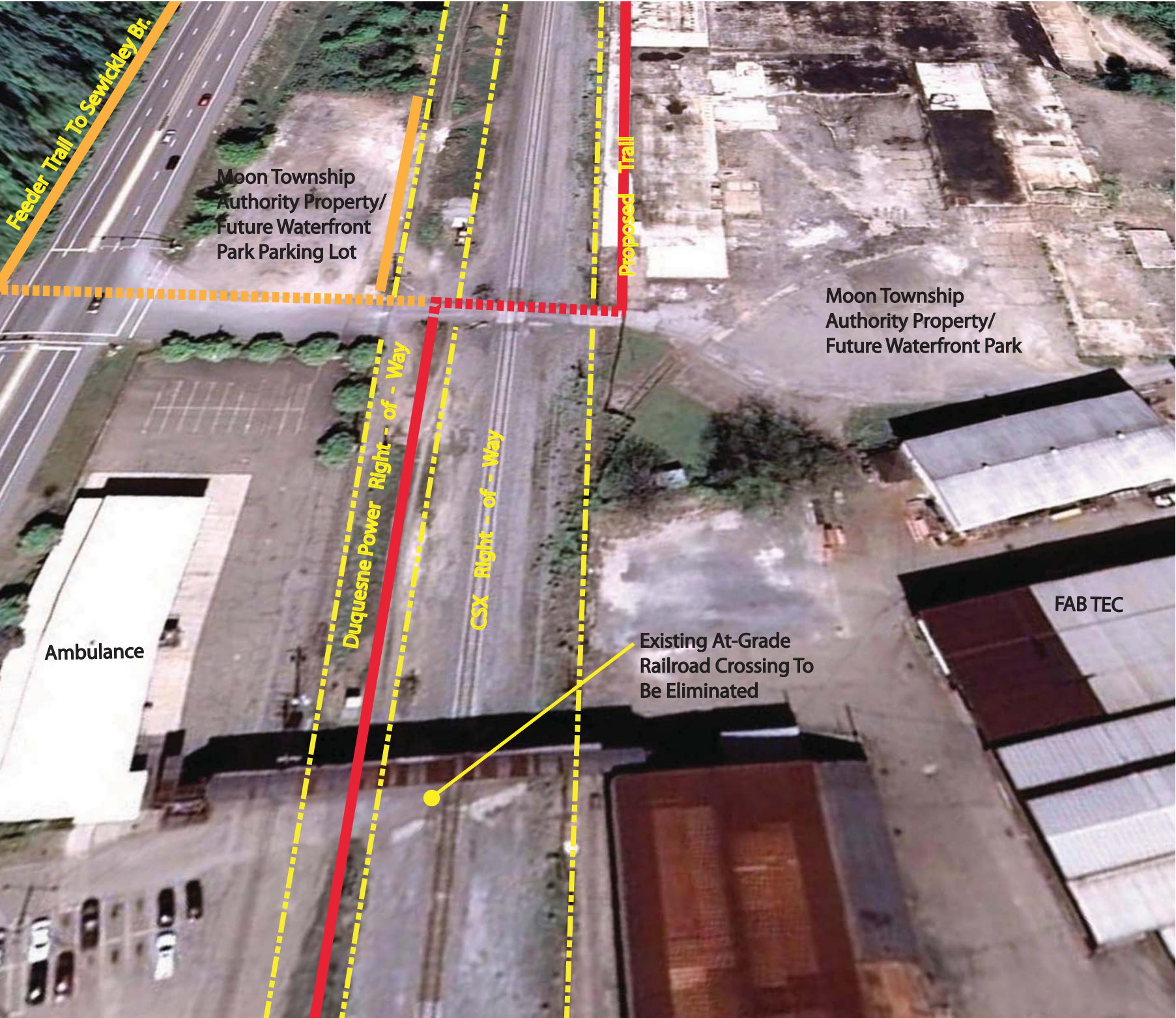


Site Photo Location

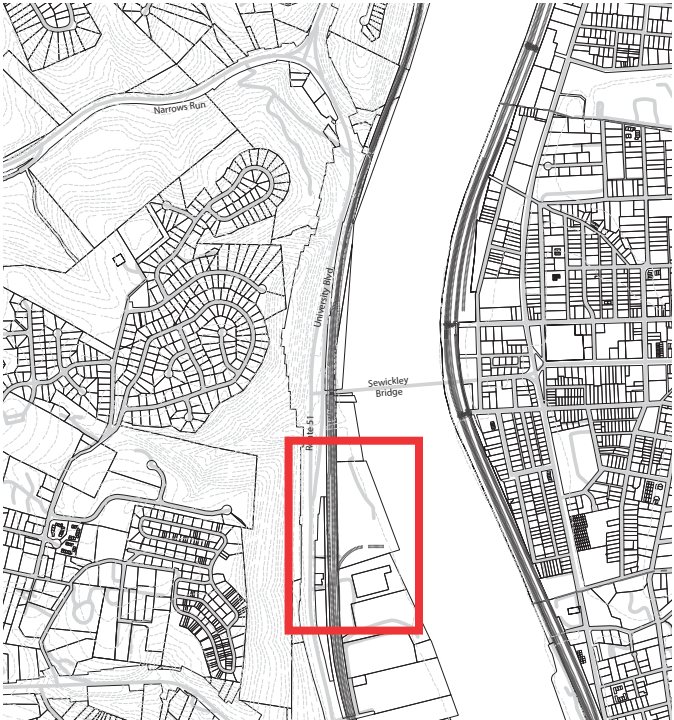
521 Proposed Trail Location
SCALE: Not to Scale

LEGEND

- Proposed 10' Wide Multi-Use Path Trail Route
- Proposed On-Road Signed Trail Route
- Neighborhood Feeder 10' Wide Side Path Trail Route
- Neighborhood Feeder On-Road Signed Trail Route



Existing At-Grade Crossing at Valley Ambulance Drive

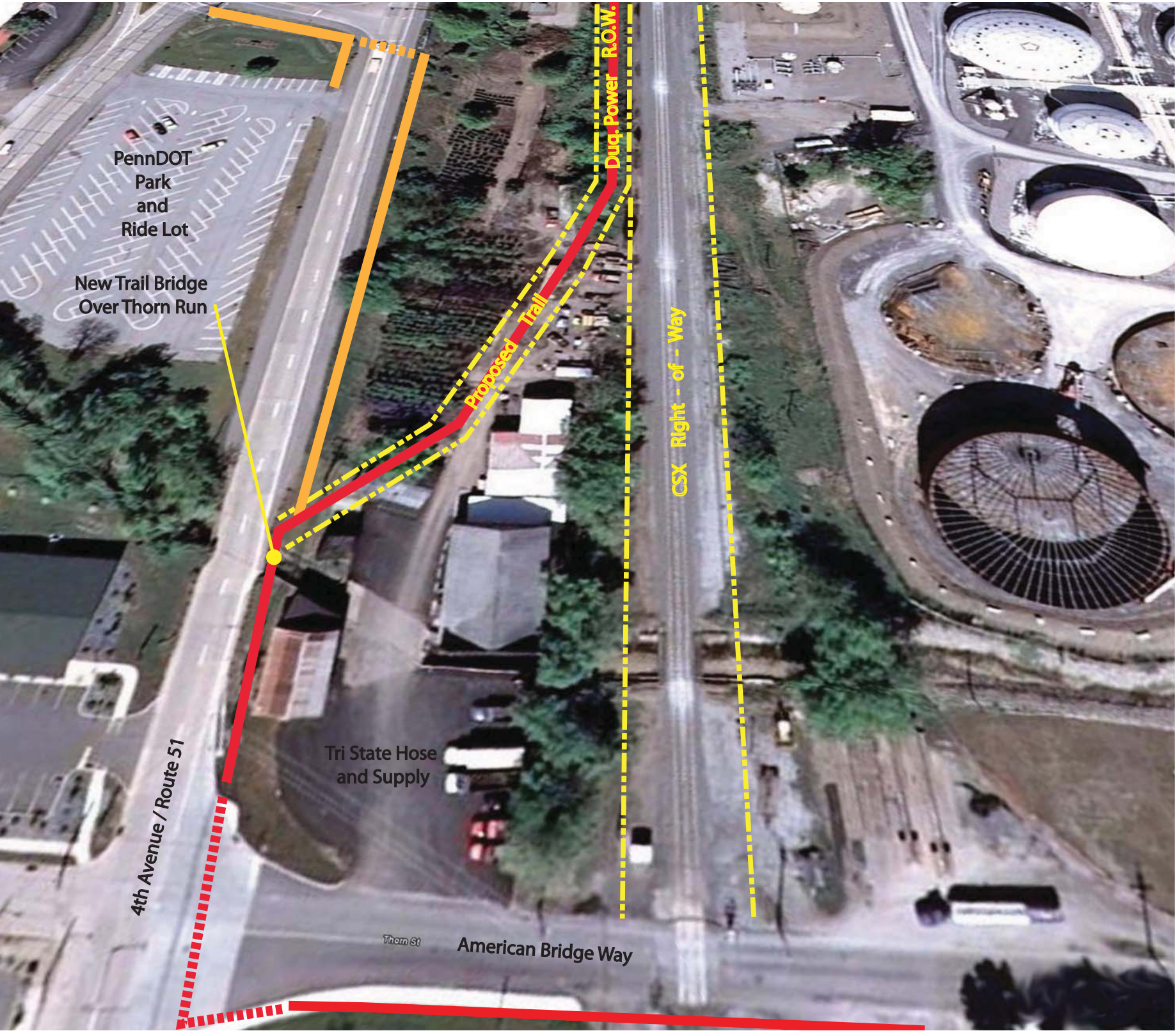


Site Photo Location

521 Proposed Trail Location
SCALE: Not to Scale

LEGEND

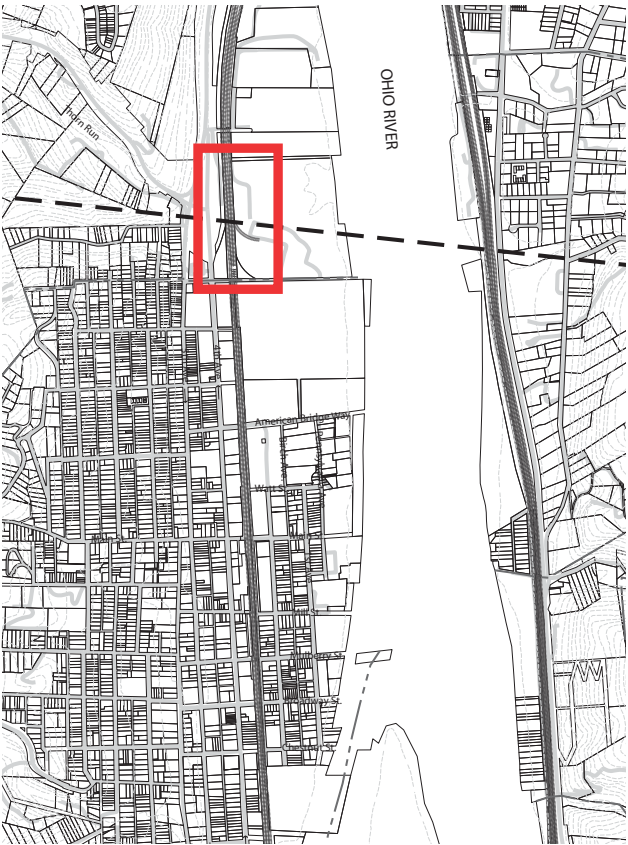
- Proposed 10' Wide Multi-Use Path Trail Route
- Proposed On-Road Signed Trail Route
- Neighborhood Feeder 10' Wide Side Path Trail Route
- Neighborhood Feeder On-Road Signed Trail Route



523 Proposed Trail Location
SCALE: Not to Scale



PennDOT Right-of-Way at Thorn Run



Site Photo Location